

when the purple-flower factor was present. In Mendel’s terminology, purple flower color is a *dominant* trait, and white flower color is a *recessive* trait. The reappearance of white-flowered plants in the F₂ generation was evidence that the heritable factor causing white flowers had not been diluted or destroyed by coexisting with the purple-flower factor in the F₁ hybrids. Mendel observed the same pattern of inheritance in six other characters, each represented by two distinctly different traits (Table 14.1). For example, when Mendel crossed a true-breeding variety that produced smooth, round pea seeds with one that produced wrinkled seeds, all the F₁ hybrids produced round seeds; this is the dominant trait for seed shape. In the F₂ generation, approximately 75% of the seeds were round and 25% were wrinkled—a 3:1 ratio, as in Figure 14.3. Now let’s see how Mendel deduced the law of segregation from his experimental results. In the discussion that follows, we will use modern terms instead of some of the terms used by Mendel. (For example, we’ll use “gene” instead of Mendel’s “heritable factor.”)

Table 14.1 The Results of Mendel’s F ₁ Crosses for Seven Characters in Pea Plants					
Character	Dominant Trait	×	Recessive Trait	F ₂ Generation Dominant: Recessive	Ratio
Flower color	Purple 	×	White 	705:224	3.15:1
Seed color	Yellow 	×	Green 	6,022:2,001	3.01:1
Seed shape	Round 	×	Wrinkled 	5,474:1,850	2.96:1
Pod color	Green 	×	Yellow 	428:152	2.82:1
Pod shape	Inflated 	×	Constricted 	882:299	2.95:1
Flower position	Axial 	×	Terminal 	651:207	3.14:1
Stem length	Tall 	×	Dwarf 	787:277	2.84:1

Mendel’s Model

Mendel developed a model to explain the 3:1 inheritance pattern that he consistently observed among the F₂ offspring in his pea experiments. We’ll now explore four related concepts making up Mendel’s model, the fourth of which is the law of segregation.

First, *alternative versions of genes account for variations in inherited characters*. The gene for flower color in pea plants, for example, exists in two versions, one for purple flowers and the other for white flowers. These alternative versions of a gene are called **alleles**. Today, we can relate this concept to chromosomes and DNA. As shown in Figure 14.4, each gene is a sequence of nucleotides at a specific place, or locus, along a particular chromosome. The DNA at that locus, however, can vary slightly in its nucleotide sequence. This variation in information content can affect the function of the encoded protein and thus an inherited character of the organism. The purple-flower allele and the white-flower allele are two DNA sequence variations possible at the flower-color locus on a pea plant’s chromosomes. The purple-flower allele sequence allows synthesis of purple pigment, and the white-flower allele sequence does not.

Second, *for each character, an organism inherits two versions (that is, two alleles) of a gene, one from each parent*. Remarkably, Mendel made this deduction without knowing about the role, or even the existence, of chromosomes. Each somatic cell in a diploid organism has two sets of chromosomes, one set inherited from each parent (see Figure 13.4). Thus, a genetic locus is actually represented twice in a diploid cell, once on each homolog of a specific pair of chromosomes. The two alleles at a particular locus may be identical, as in the true-breeding plants of Mendel’s P generation. Or the alleles may differ, as in the F₁ hybrids (see Figure 14.4).

Third, *if the two alleles at a locus differ, then one, the **dominant allele**, determines the organism’s appearance; the other, the **recessive allele**, has no noticeable effect on the organism’s appearance*. Accordingly, Mendel’s F₁ plants had purple flowers because the allele for that trait is dominant and the allele for white flowers is recessive.

The fourth and final part of Mendel’s model, the **law of segregation**, states that *the two alleles for a heritable character segregate (in other words, separate from each other) during gamete formation and end up in different gametes*. Thus, an egg or a sperm gets only one of the two alleles that are present in the diploid cells of the organism making the gamete. In terms of chromosomes, this segregation corresponds to the distribution of copies of the two members of a pair of homologous chromosomes to different gametes in meiosis (see Figure 13.7). Note that if an organism has identical alleles for a particular character, then that allele is present in all gametes. Because it is the only allele that can be passed on to offspring when the plant self-pollinates, the offspring always look the same as their parents in regard to that characteristic; this explains why these plants are true-breeding. But if different alleles are